

Mechanical Engineering	Electronics, Communication and Electrical Engineering	Instrumentation Technology	Control and Automation
Manufacturing and Production	Automotive Infotonics and Informatics		Experts from Automotive Industries
Applied Physics and Nanotechnology			Mechatronics and Robotics
Management			Civil and Transportation Management
Environmental Engineering	Computer Science and IT	Energy Engineering	Professional bodies like Society of Automotive Engineers

Fig. 1: Interdisciplinarity of automotive domain in various branches of engineering

had to be done according to the new place. This record-keeping management technique became a good way to collect revenues or taxes only from those who use automobiles rather than collecting it from all citizens. This revenue, hence, was used as investment for infrastructure like roads for speed, comfort and reliability of users.

Risk of losing such possessions and paying to buy these time and again is checked by a well-established insurance department, which requires accurate records of both automobiles and their operators, along with their histories. Interaction of these records has made it possible to construct an ongoing structure of rewards and sanctions for operators, thereby offering premium discounts to good drivers, whereas giving higher premium or no service at all to others.

In addition to this, maintaining a proper registry system of all crimes and criminals related to automobiles by the crime department makes prosecution of criminals easier and faster by just going through the criminal records.

Rise of extensive record keeping related to automobiles has resulted in time-series databases that facilitates systematic and scientific analysis of many issues. Gradually, new technologies such as electromechanical unit record equipment and digital computers came into the picture, proving to be better than paper based records. It is IT that is responsible for all stored

data to be highly-secure, accurate and reliable.

Pollution and emission control to combat climate change. Pollution due to growing population of automobiles is of major concern in the last two decades, as harmful gases like carbon monoxide, oxides of nitrogen, sulphur oxides are emitted. Researchers demonstrated in 1956 that smog was manufactured in the air by the action of sunlight on chemicals in the exhaust of automobiles.

A number of advanced technologies were adopted to get through the problem of foreign competition versus original equipment manufacturers (OEMs), but no breakthrough happened until the development of a three-way catalytic convertor that used rare-earth metals as catalysts and converted CO into CO₂, hydrocarbons into CO₂ and water and oxides of nitrogen into N₂.

Although it converted 98 per cent of harmful gases into harmless ones, the convertor was very sensitive to stoichiometric conditions. The solution to this was Lambda-Sond System developed by Volvo in 1970. The system introduced the use of a digital sensor at the junction of the exhaust and catalytic convertor, thereby detecting the slightest disturbance in stoichiometric ratios.

This was the first application of modern IT to control automobiles for emission control. This concept was used in Volvo's 240 model launched

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